

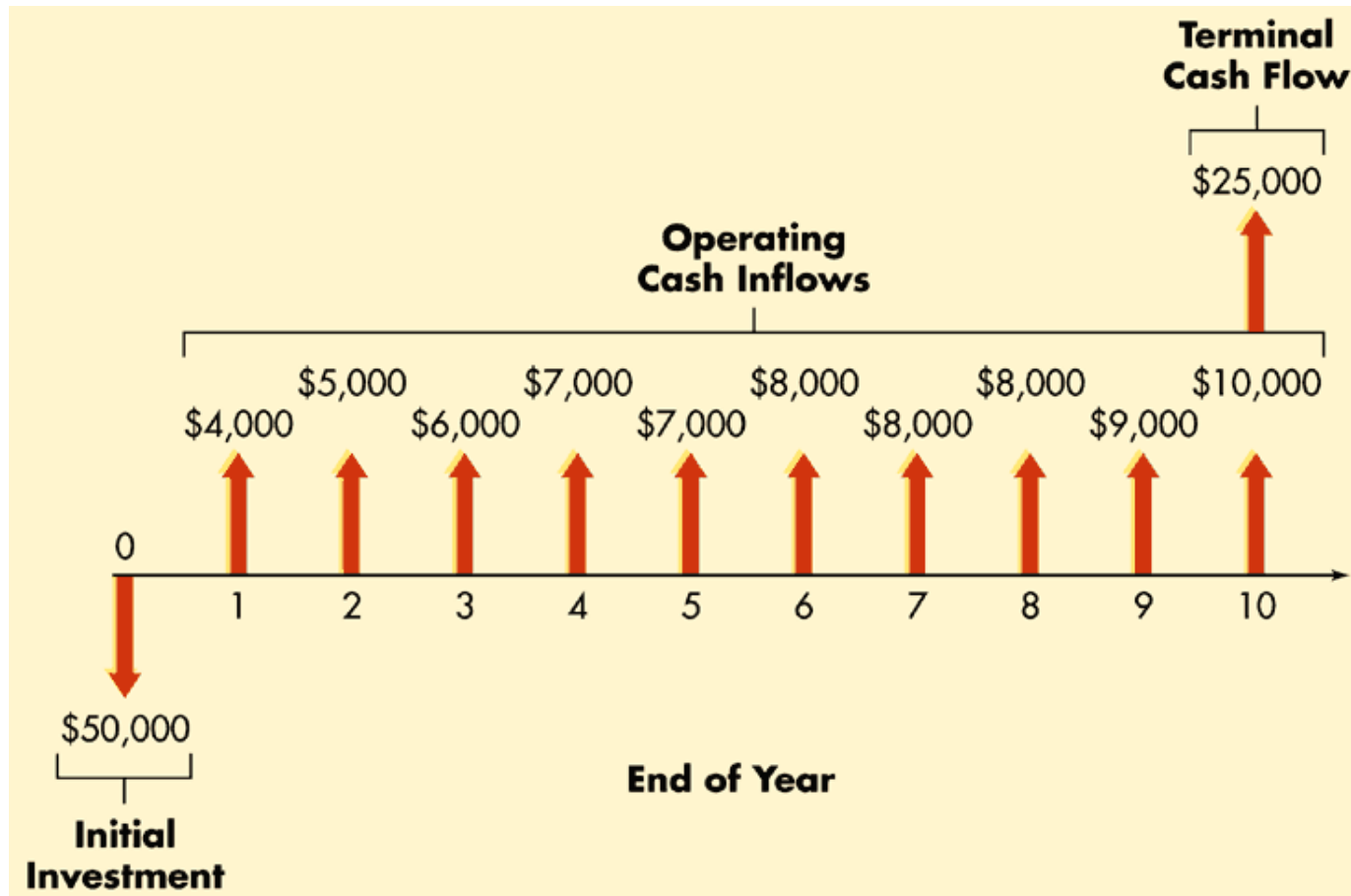
Summer Term 2026

Investment and Financial Management

Sessions 03+04: Investment Analysis

Berk/DeMarzo (2024): Corporate Finance, 6th edition, Pearson: Chapter 7

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- 1. The net present value (NPV)**
 - 2. The internal rate of return (IRR)**
 - 3. Other methods**
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Source: Gitman, L.J. and Zutter, C.J., 2015
(14th edition), Principles of Managerial Finance.

1. The net present value (NPV)

2. The internal rate of return (IRR)

3. Other methods

■ Definition of the net present value (NPV)

The net present value (NPV) of a project or investment is the difference between the present value of its benefits and the present value of its costs.

$$\text{NPV} = \text{PV}(\text{Benefits}) - \text{PV}(\text{Costs}) = \text{PV}(\text{All project cash flows})$$

$$\text{NPV} = \sum_{t=0}^T \frac{\text{CF}_t}{(1+r)^t}$$

With:

NPV = Net present value

PV = Present value

CF_t = Cash flow in period t

r = Appropriate discount rate

1. The net present value (NPV)

▪ The net present value (NPV) rule

NPV decision rule: When making an investment decision, take the alternative with the highest NPV. Choosing this alternative is equivalent to receiving its NPV in cash today.

Accepting or rejecting a project:

- *Accept those projects with positive NPV because accepting them is equivalent to receiving their NPV in cash today!*
- *Reject those projects with negative NPV because accepting them would reduce the wealth of investors!*

1. The net present value (NPV)

- **The net present value (NPV) rule**

The NPV Is Equivalent to Cash Today

Problem

Your firm needs to buy a new \$9500 copier. As part of a promotion, the manufacturer has offered to let you pay \$10,000 in one year, rather than pay cash today. Suppose the risk-free interest rate is 7% per year. Is this offer a good deal? Show that its NPV represents cash in your pocket.

1. The net present value (NPV)

■ The net present value (NPV) rule

Solution

If you take the offer, the benefit is that you won't have to pay \$9500 today, which is already in PV terms. The cost, however, is \$10,000 in one year. We therefore convert the cost to a present value at the risk-free interest rate:

$$PV(\text{Cost}) = (\$10,000 \text{ in one year}) \div (1.07 \$ \text{ in one year} / \$ \text{ today}) = \$9345.79 \text{ today}$$

The NPV of the promotional offer is the difference between the benefits and the costs:

$$NPV = \$9500 - \$9345.79 = \$154.21 \text{ today}$$

The NPV is positive, so the investment is a good deal. It is equivalent to getting a cash discount today of \$154.21, and only paying \$9345.79 today for the copier. To confirm our calculation, suppose you take the offer and invest \$9345.79 in a bank paying 7% interest. With interest, this amount will grow to $\$9345.79 \times 1.07 = \$10,000$ in one year, which you can use to pay for the copier.

- **Another example: Net present value (NPV) rule**

Problem

Consider a take-it-or-leave-it investment decision involving a single, stand-alone project for Fredrick's Feed and Farm (FFF). The project costs \$250 million and is expected to generate cash flows of \$35 million per year, starting at the end of the first year and lasting forever.

1. The net present value (NPV)

- **Another example: Net present value (NPV) rule**

Solution (I)

The NPV of the project is calculated as:

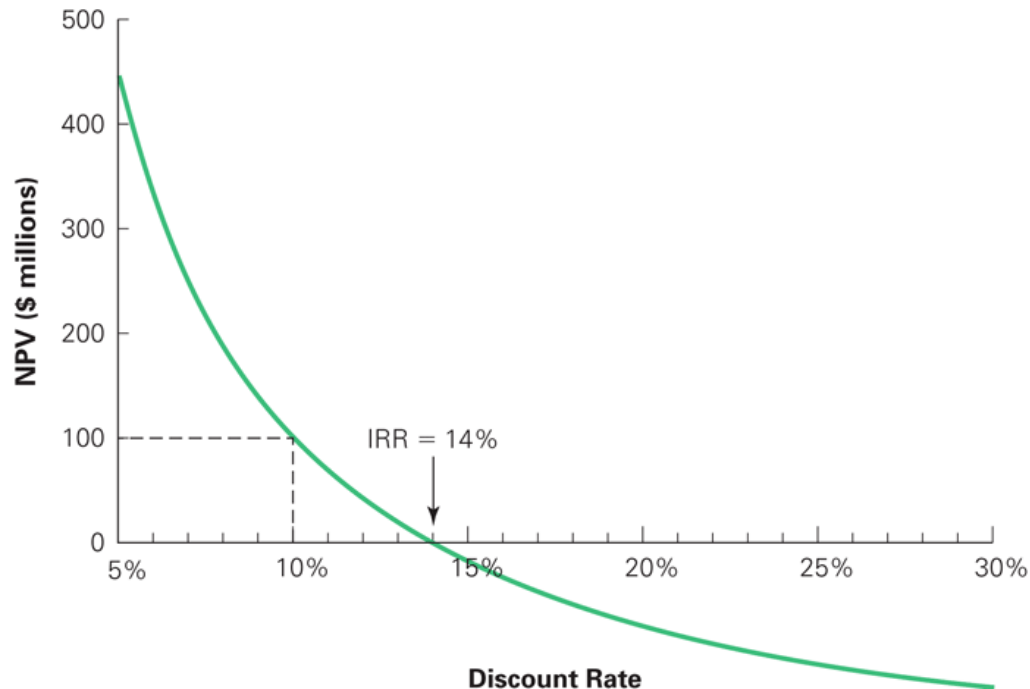
$$\text{NPV} = -250 + \frac{35}{r}$$

The NPV is dependent on the discount rate.

1. The net present value (NPV)

- Another example : Net present value (NPV) rule

Solution (II)



Note:

If FFF's cost of capital is 10%, the NPV is \$100 million and they should undertake the investment.

▪ Net present value (NPV) rule – A summary

- *Conduct only projects with $NPV > 0$*
- *When choosing among alternative projects, select the one with the highest NPV. This alternative is the one which will lead to the largest increase in the value of the firm.*
- *Sometimes alternative investment rules may give the same answer as the NPV rule, but at other times they may disagree.*

When the rules conflict, the NPV decision rule should be followed!

1. The net present value (NPV)

2. The internal rate of return (IRR)

3. Other methods

2. The internal rate of return (IRR)

- **Definition of the internal rate of return (IRR)**

The internal rate of return (IRR) is the discount rate that makes the net present value (NPV) equal to zero.

$$NPV = \sum_{t=0}^N \frac{CF_t}{(1+IRR)^t} = 0$$

With:

NPV = Net present value

CF_t = Cash flow in period t

IRR = Internal rate of return

- **Internal rate of return (IRR) investment rule**

IRR investment rule: Take any investment where the IRR exceeds the cost of capital. Turn down any investment whose IRR is less than the cost of capital.

- **Application of the IRR rule: IRR vs. NPV**

The IRR rule works for a stand-alone project if all of the project's negative cash flows precede its positive cash flows (= “normal” cash flow projects). ➔ In other cases, the IRR rule may disagree with the NPV rule and thus be incorrect.

- **Pitfalls of the IRR rule**

Situations where the IRR rule and NPV rule may be in conflict:

- Pitfall #1: Delayed investments
- Pitfall #2: Multiple IRRs
- Pitfall #3: Nonexistent IRR

- **Pitfall #1: Delayed investments**

Problem

Assume you have just retired as the CEO of a successful company. A major publisher has offered you a book deal. The publisher will pay you \$1 million upfront if you agree to write a book about your experiences. You estimate that it will take three years to write the book. The time you spend writing will cause you to give up speaking engagements amounting to \$500,000 per year. You estimate your opportunity cost to be 10%.

Should you accept the deal?

2. The internal rate of return (IRR)

- **Pitfall #1: Delayed investments**

Solution (I)

1. Calculate the IRR (e.g., in Excel)

	NPER	RATE	PV	PMT	FV	Excel Formula
Given	3		1,000,000	−500,000	0	
Solve for I		23.38%				RATE(3, 500000, 1000000, 0)

The IRR is greater than the cost of capital. Thus, the IRR rule indicates you should accept the deal.

2. Calculate the NPV

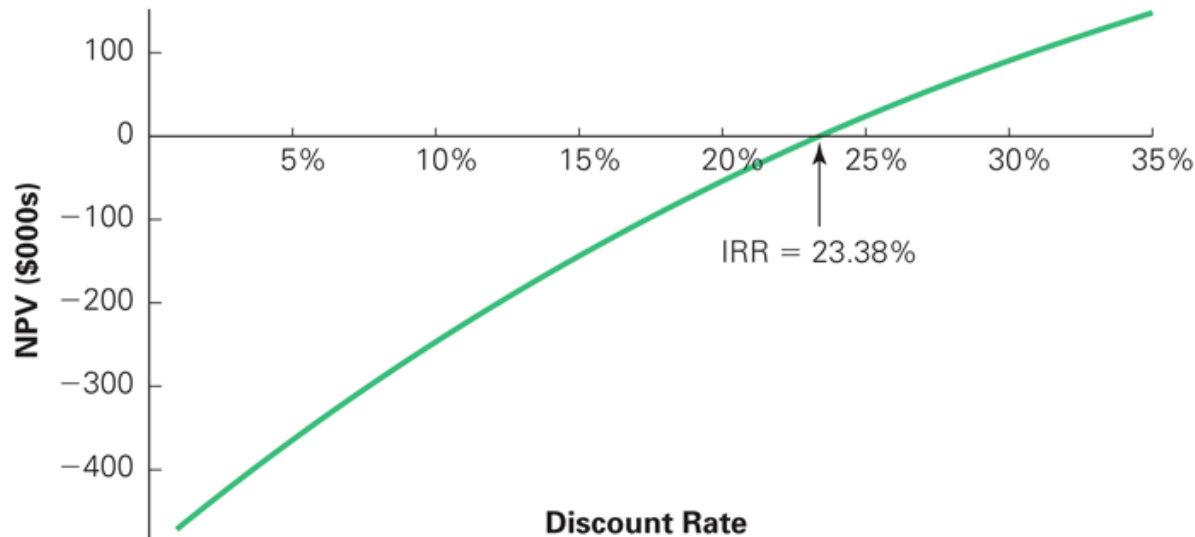
$$\text{NPV} = 1,000,000 - \frac{500,000}{1.1} - \frac{500,000}{1.1^2} - \frac{500,000}{1.1^3} = -\$243,426$$

Since the NPV is negative, the NPV rule indicates you should reject the deal.

2. The internal rate of return (IRR)

- **Pitfall #1: Delayed investments**

Solution (II)



Note:

When the benefits of an investment occur before the costs, the NPV is an increasing function of the discount rate.

- **Pitfall #2: Multiple IRRs**

Problem

Suppose Star informs the publisher that it needs to sweeten the deal before he will accept it. The publisher offers \$550,000 advance and \$1,000,000 in four years when the book is published.

Should he accept or reject the new offer?

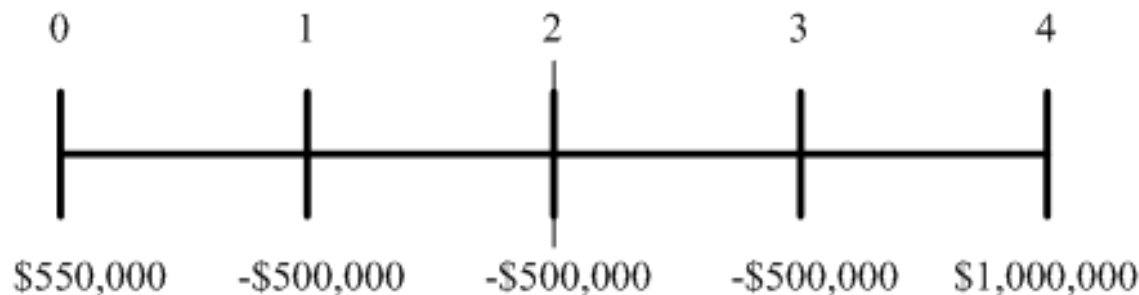
2. The internal rate of return (IRR)

- **Pitfall #2: Multiple IRRs**

Solution (I)

1. Calculate the IRR (e.g., in Excel)

The cash flows would now look like:



Note that CFs are non-normal, as their sign changes twice:

$$NPV = 550,000 - \frac{500,000}{(1+r)} - \frac{500,000}{(1+r)^2} - \frac{500,000}{(1+r)^3} + \frac{1,000,000}{(1+r)^4}$$

- **Pitfall #2: Multiple IRRs**

Solution (II)

By setting the NPV equal to zero and solving for r , we find the IRR. In this case, there are two IRRs: **7.164%** and **33.673%**.

Because there is more than one IRR, the IRR rule cannot be applied.

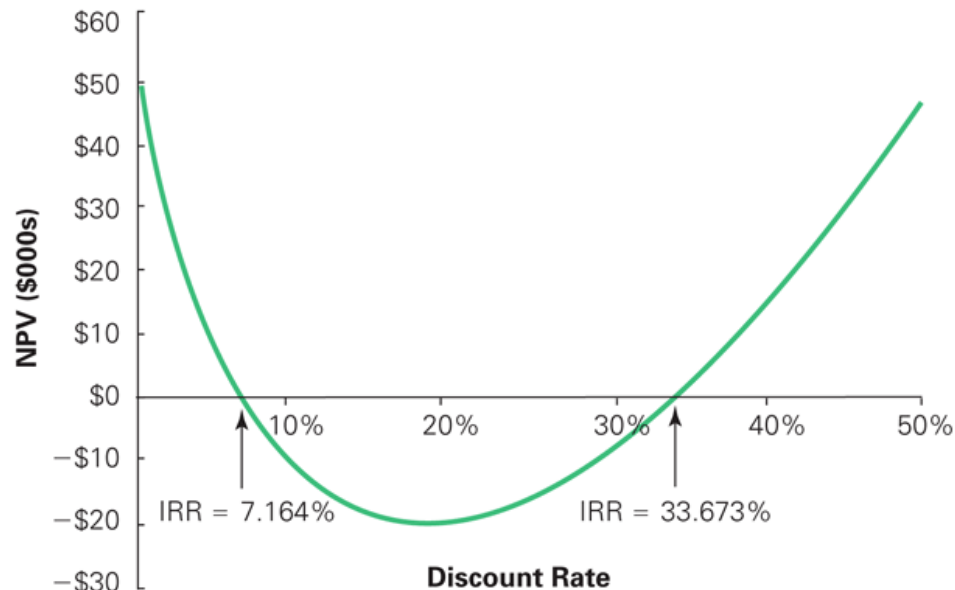
2. Calculate the NPV

Between 7.164% and 33.673%, the book deal has a negative NPV. Since your opportunity cost of capital is 10%, you should reject the deal!

2. The internal rate of return (IRR)

- **Pitfall #2: Multiple IRRs**

Solution (III)



- **Pitfall #3: Nonexistent IRR**

Problem

Finally, Star is able to get the publisher to increase his advance to \$750,000, in addition to the \$1 million when the book is published in four years.

Should you accept the deal?

- **Pitfall #3: Nonexistent IRR**

Solution (I)

1. Calculate the IRR (e.g., in Excel)

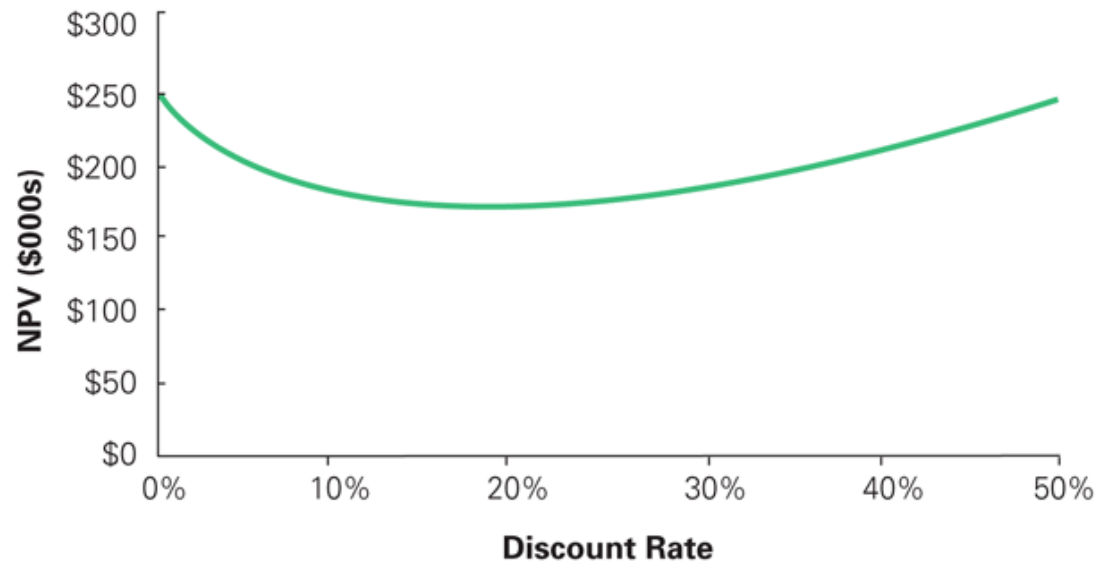
$$\text{NPV} = 750,000 - \frac{500,000}{(1+r)} - \frac{500,000}{(1+r)^2} - \frac{500,000}{(1+r)^3} + \frac{1,000,000}{(1+r)^4}$$

With these cash flows, no IRR exists; there is no discount rate that makes the NPV equal to zero.

2. The internal rate of return (IRR)

- **Pitfall #3: Nonexistent IRR**

Solution (II)



Note:

No IRR exists because the NPV is positive for all values of the discount rate. Thus, the IRR rule cannot be used.

2. The internal rate of return (IRR)

- **IRR versus the IRR rule**

While the IRR rule has shortcomings for making investment decisions, the IRR itself remains useful.

- **But:**

- IRR measures the average return of the investment. Hence, it is an exact measure for the average ROIC of a project over its lifetime.
- IRR can be used to check the sensitivity of the NPV to any estimation error in the cost of capital.

2. The internal rate of return (IRR)

- IRR versus the IRR rule

Problems with the IRR Rule

Problem

Consider projects with the following cash flows:

Project	0	1	2
A	−375	−300	900
B	−22,222	50,000	−28,000
C	400	400	−1,056
D	−4,300	10,000	−6,000

Which of these projects have an IRR close to 20%? For which of these projects is the IRR rule valid?

- **IRR versus the IRR rule**

Solution

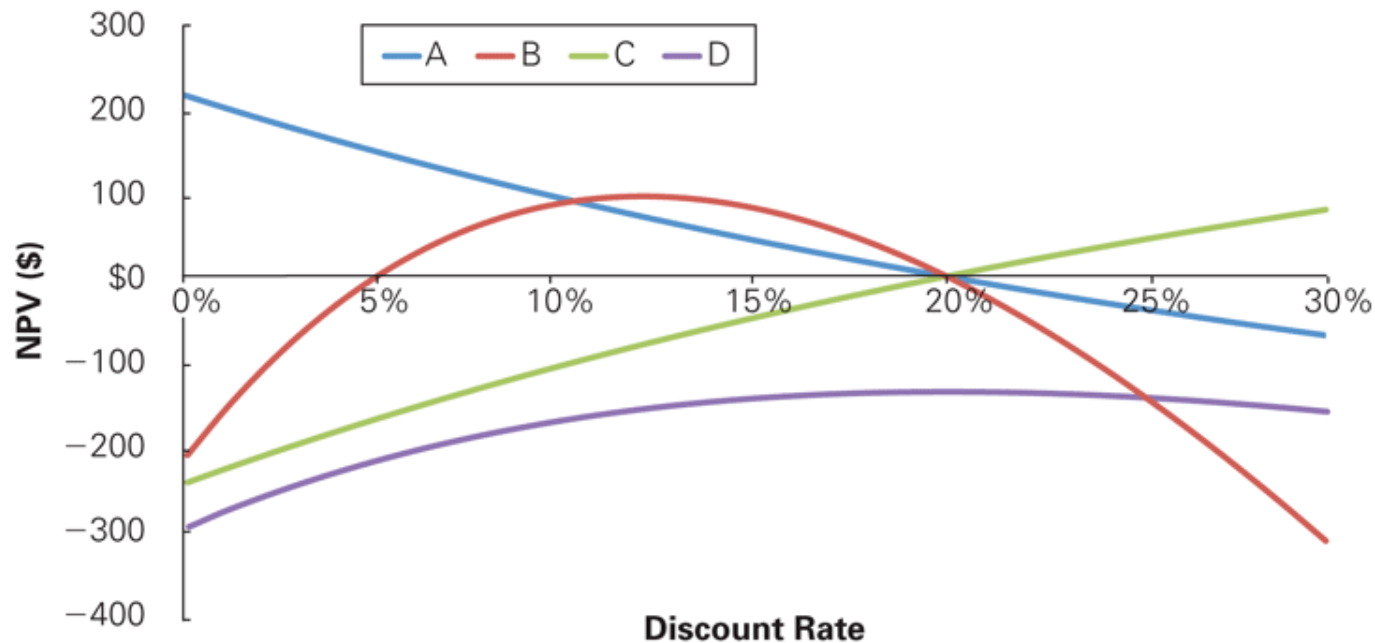
We plot the NPV profile for each project in Figure 7.5. From the NPV profiles, we can see that projects A, B, and C each have an IRR of approximately 20%, while project D has no IRR. Note also that project B has another IRR of 5%.

The IRR rule is valid only if the project has a positive NPV for every discount rate below the IRR. Thus, the IRR rule is only valid for project A. This project is the only one for which all the negative cash flows precede the positive ones.

2. The internal rate of return (IRR)

- **IRR versus the IRR rule**

While the IRR Rule works for project A, it fails for each of the other projects.



2. The internal rate of return (IRR)

▪ Mutually exclusive projects

- When you must choose only one project among several possible projects, the choice is mutually exclusive.

NPV rule: Select the project with the highest NPV.

IRR rule: Selecting the project with the highest IRR may lead to mistakes.

2. The internal rate of return (IRR)

- **Project ranking with different scales:** As the IRR is a rate of return, one cannot compare projects of different scale (size).

NPV and Mutually Exclusive Projects

Problem

A small commercial property is for sale near your university. Given its location, you believe a student-oriented business would be very successful there. You have researched several possibilities and come up with the following cash flow estimates (including the cost of purchasing the property). Which investment should you choose?

Project	Initial Investment	First-Year Cash Flow	Growth Rate	Cost of Capital
Book Store	\$300,000	\$63,000	3.0%	8%
Coffee Shop	\$400,000	\$80,000	3.0%	8%
Music Store	\$400,000	\$104,000	0.0%	8%
Electronics Store	\$400,000	\$100,000	3.0%	11%

- **Project ranking with different scales**

Solution

Assuming each business lasts indefinitely, we can compute the present value of the cash flows from each as a constant growth perpetuity. The NPV of each project is

$$NPV(\text{Book Store}) = -300,000 + \frac{63,000}{8\% - 3\%} = \$960,000$$

$$NPV(\text{Coffee Shop}) = -400,000 + \frac{80,000}{8\% - 3\%} = \$1,200,000$$

$$NPV(\text{Music Store}) = -400,000 + \frac{104,000}{8\%} = \$900,000$$

$$NPV(\text{Electronics Store}) = -400,000 + \frac{100,000}{11\% - 3\%} = \$850,000$$

Thus, all of the alternatives have a positive NPV. But, because we can only choose one, the coffee shop is the best alternative.

2. The internal rate of return (IRR)

- **IRR rule and mutually exclusive investment with project ranking with different scales:**
 - If a project's size is doubled, its NPV will double. This is not the case with IRR. Thus, the IRR rule cannot be used to compare projects of different scales.
 - Consider two of the projects from our given example.

	Bookstore	Coffee shop
Initial investment	\$300,000	\$400,000
Cash flow (in year 1)	\$63,000	\$80,000
Annual growth rate	3%	3%
Cost of capital	8%	8%
IRR	24%	23%
NPV	\$960,000	\$1,200,000

■ Project ranking with timing of cash flows

- Another problem with the IRR is that it can be affected by changing the timing of the cash flows, even when the scale is the same.
- IRR is a return, but the dollar value of earning (a given return) depends on how long the return is earned.
- Consider again the coffee shop and the music store investment in our example. Both have the same initial scale and the same horizon. The coffee shop has a lower IRR, but a higher NPV because of its higher growth rate.

▪ Project ranking with differences in risk

- An IRR that is attractive for a safe project need not be attractive for a riskier project.
- Consider the investment in the electronics store from our example. The IRR is higher than those of the other investment opportunities, yet the NPV is the lowest.
- The higher cost of capital means a higher IRR is necessary to make the project attractive.

1. The Net Present Value (NPV)

2. The Internal Rate of Return (IRR)

3. Other methods

- **Definition of the payback rule**

The payback period is amount of time it takes to recover or pay back the initial investment.

- **Payback rule**

Payback rule: If the payback period is less than a pre-specified length of time, you accept the project. Otherwise, you reject the project!

- **Applicability of payback rule**

The payback rule is used by many companies because of its simplicity. This is a widespread method in practice. Maybe, because firms often care more about the liquidity drain of a project rather than its profitability.

- **Payback rule**

Problem

Projects A, B, and C each have an expected life of 5 years.

Given the initial cost and annual cash flow information below, what is the payback period for each project?

	Project A	Project B	Project C
Cost	\$80	\$120	\$150
Cash flow	\$25	\$30	\$35

▪ Payback rule

Solution

- **Project A**
 $\$80 \div \$25 = 3.2 \text{ years}$
- **Project B**
 $\$120 \div \$30 = 4.0 \text{ years}$
- **Project C**
 $\$150 \div \$35 = 4.29 \text{ years}$

- **Shortcomings of the payback rule**

- Ignores the project's cost of capital and time value of money
- Ignores cash flows after the payback period
- Relies on an ad hoc decision criterion

- **Definition of the profitability index (PI)**

The profitability index can be used to identify the optimal combination of projects to undertake. Companies use it for the evaluation of projects with different resource constraints.

$$\text{Profitability index} = \frac{\text{Value created}}{\text{Resource consumed}} = \frac{\text{Net present value}}{\text{Resource consumed}}$$

- **Profitability index (PI) rule**

PI rule: When choosing among projects competing for the same resource, rank the projects by their profitability indices and pick the set of projects with the highest profitability indices that can still be undertaken given the limited resource.

- **Application of the profitability index (PI)**

Profitability Index with a Human Resource Constraint

Problem

Your division at NetIt, a large networking company, has put together a project proposal to develop a new home networking router. The expected NPV of the project is \$17.7 million, and the project will require 50 software engineers. NetIt has a total of 190 engineers available, and the router project must compete with the following other projects for these engineers:

Project	NPV (\$ millions)	Engineering Headcount
Router	17.7	50
Project A	22.7	47
Project B	8.1	44
Project C	14.0	40
Project D	11.5	61
Project E	20.6	58
Project F	12.9	32
Total	107.5	332

How should NetIt prioritize these projects?

■ Application of the profitability index (PI)

Solution

The goal is to maximize the total NPV we can create with 190 engineers (at most). We compute the profitability index for each project, using Engineering Headcount in the denominator, and then sort projects based on the index:

Project	NPV (\$ millions)	Engineering Headcount (EHC)	Profitability Index (NPV per EHC)	Cumulative EHC Required
Project A	22.7	47	0.483	47
Project F	12.9	32	0.403	79
Project E	20.6	58	0.355	137
Router	17.7	50	0.354	187
Project C	14.0	40	0.350	
Project D	11.5	61	0.189	
Project B	8.1	44	0.184	

We now assign the resource to the projects in descending order according to the profitability index. The final column shows the cumulative use of the resource as each project is taken on until the resource is used up. To maximize NPV within the constraint of 190 engineers, NetIt should choose the first four projects on the list. There is no other combination of projects that will create more value without using more engineers than we have. Note, however, that the resource constraint forces NetIt to forgo three otherwise valuable projects (C, D, and B) with a total NPV of \$33.6 million.

▪ Shortcomings of the profitability index (PI)

- In some situations the profitability index does not give an accurate answer:
 - Suppose in our example that NetIt has an additional small project with a NPV of only \$120,000 that requires 3 engineers. The profitability index in this case is $0.12 / 3 = 0.04$, so this project would appear at the bottom of the ranking. However, 3 of the 190 employees are not being used after the first four projects are selected. As a result, it would make sense to take on this project even though it would be ranked last.
- One would have to enumerate the different combinations in order to find out the NPV maximizing combination.
- With multiple resource constraints, the profitability index can break down completely.

■ Chapter quiz of session 03 + 04

1. Explain the NPV rule for stand-alone projects.
2. If the IRR rule and the NPV rule lead to different decisions for a stand-alone project, which should you follow? Why?
3. For mutually exclusive projects, explain why picking one project over another because it has a larger IRR can lead to mistakes.
4. Explain why ranking projects according to their NPV might not be optimal when you evaluate projects with different resource requirements.
5. How can the profitability index be used to identify attractive projects when there are resource constraints?